Budapest University of Technology and Economics

Faculty of Natural Sciences

M2 - Basic connections for operational amplifiers

Lab report

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Abstract

This work presents the results obtained from the laboratory session focusing on the operation of different amplifier configurations including the inverting amplifier, adder amplifier, non-inverting voltage follower, differentiator and integrator. The voltage gain, saturation and upper cutoff frequency are determined experimentally. The differentiator and integrator circuits were studied using sinusoidal, square and triangular input waveforms, and the effects of additional feedback components were investigated.

1 Theoretical introduction

This measurement focused on the operational amplifiers (opamps). Different configurations of these opamps, result in various applications, such as a voltage or current amplifier, both inverting and non inverting modes, voltage follower (also known as buffer), adder and subtractor, and integration and differentiator circuits.

The basic operation of an opamp relies on the potential difference between its inputs. This potential difference is multiplied by some gain factor to result in the output voltage. It follows the equation:

$$U_{out} = A_u \cdot (U_1 - U_2) \tag{1}$$

With U_1 and U_2 being the voltages at the non-inverting and inverting inputs, respectively.

The basic configurations of the opamp mentioned above rely on the basis of being inverting or non-inverting. The differentiation between the two modes is the input signal of the opamp. If the input voltage is fed into the negative (inverting) input then the input signal will be negatively multiplied resulting in an output of the opposite sign. On the non-inverting (positive) input, the sign of the output voltage will be analogous to that of the input voltage. The amplitude of the opamp is configured by the ratios of the resistances of the circuit, with a resistor on the inverting input R_s and a feedback resistor R_v.

Amplification of inverting amplifier:

$$A_u = -\left(\frac{R_v}{R_s}\right) \tag{2}$$

Amplification of non-inverting amplifier:

$$A_u = 1 + \left(\frac{R_v}{R_s}\right) \tag{3}$$

The opamp requires a power supply. Upon saturation, when the opamp reaches its absolute maximum amplification gain, the output voltage saturates to the positive or negative supply voltage, depending on the circuitry being an inverting or non-inverting amplifier. There is a

linear region of amplification before the saturation, in this region the output directly follows the input differential voltage equation shown above. However, upon saturation this linearity ceases, the output voltage caps, it shows a cut off when the input differential exceeds the threshold limit, this is what we refer to as the saturation of the signal.

Often we calculate with ideal opamps as real opamps have many factors and parasitic qualities which make calculations incredibly tiresome and very tedious, for the ideal opamps we assume an infinite input resistance and zero output resistance, with infinite bandwidth and common mode rejection, another assumption which simplifies calculation is the virtual ground assumption in the negative feedback linear mode. To near these ideal conditions, real opamps have input resistances in the MΩ magnitudes, open loop gains (amplifications) near millions, and output resistances of just a few Ωs. Our laboratory configurations don't approach these ideals as such, our setup is detailed in the section: Measurement setup below.

In the adder (summing) circuit the operation is to output a voltage that is a weighted sum of the multiple input voltages lead through the resistors, driven all through the inverting input of the opamp. In this case the node equation gives us an output voltage of:

$$\frac{U_1}{R_1} + \frac{U_2}{R_2} + \frac{U_3}{R_3} + \frac{U_{out}}{R_v} = 0 \tag{4}$$

$$U_{out} = -\left(\frac{R_v}{R_1}U_1 + \frac{R_v}{R_2}U_2 + \frac{R_v}{R_3}U_3\right) \tag{5}$$

The subtractor circuit purely uses the potential difference between the input voltages on both inverting and non-inverting input, no open loop gain or resistors affect the output voltage in the case of the subtractor circuit:

$$U_{out} = U_{in1} - U_{in2} \tag{6}$$

The voltage follower - also known as buffer - circuit has the aim of achieving unity gain, in reality we often achieve near unity gain $A_u \approx 1$. The purpose of this opamp is to provide low-impedance voltage drive to following loads from an initial high input impedance signal source.

Differentiator circuits place a capacitor between the input and feedback resistor on the amplifier, introducing an RC constant for accuracy in the equation of the output voltage, resulting in a phase shift between input and output signals. The original purpose however is to produce an output signal proportional to the rate of change of the input voltage, this protects the load from any steep changes in the input voltage, a great use in control engineering.

$$U_{out} = -CR_v\frac{dU_{in}(t)}{dt} \tag{7}$$

Integrator circuitry swaps the positions of the resistor and capacitor. This is widely used

in instruments such as radiation dose integration from pulse counts. The RC time constant appears again for accuracy in integration, resulting in the output relationship:

$$U_{out} = -\frac{1}{R_s C} \int_0^t U_{in} dt \tag{8}$$

The Miller effect appears in the high frequency range where it reduces the amplification by the parasitic stray capacitance which arises between the inverting input and output, multiplied by the stage gain:

$$C_{eff} = C\,(1 + A) \tag{9}$$

2 Measurement tools

For this measurement, we used the following list of tools:

- OK Industries Inc. CM-140 L breadboard
- ST μA741CN operational amplifier - 93d427
- Resistors: 470 kΩ (1 piece), 100 kΩ (1 piece), 1 kΩ (4 pieces), 10 MΩ (1 piece)
- Banana plugs (8 pieces)
- Crocodile clip (6 pieces)
- Jumping wires (6 pieces)
- HM 8030-4 HAMEG Function generator
- HM8040-2 HAMEG Triple power supply
- HM 8011-3 HAMEG Digital multimeter
- SIGLENT SDS 1072CML Digital storage oscilloscope
- Capacitors: 10 nF (1 piece), 100 pF (1 piece)

3 Measurement setup

We configured the circuits shown in the figures by using the breadboard. Following the hand-out's figure 4 [1] for the first task we built the inverting configuration by forming a common ground with the banana cables for all instruments, this way we could set the input voltage and measure the output voltage with respect to this common ground we established. We added a 100 kΩ resistor to the inverting input pin of the opamp (leg 2) and added the feedback resistor of 470 kΩ from leg 2 to 6 (output pin).

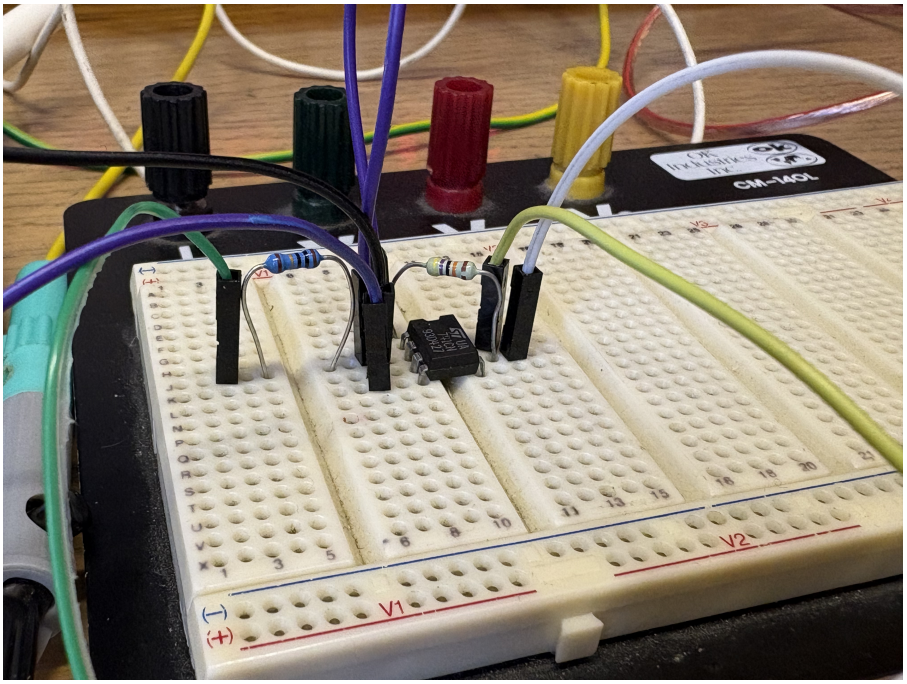


Figure 1: Inverting configuration of task 1

In task 2 we configured the adder circuit using two additional resistances on the inputs, in this setup all resistors were of the value 1 kΩ. We have to drive both input resistors by the same input voltage, because of this we used one input and connected both resistor legs to simplify the circuitry.

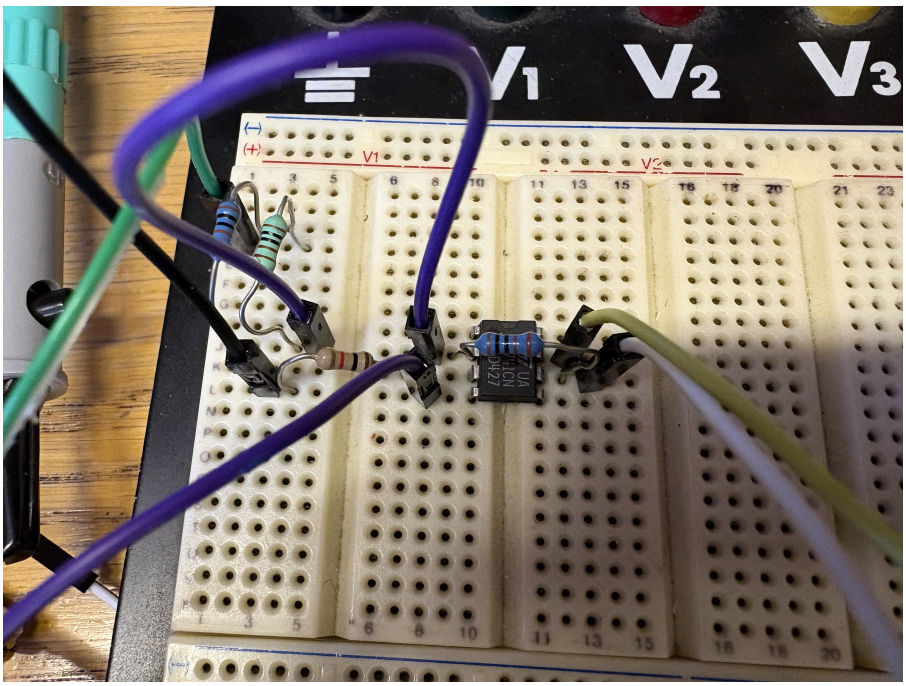


Figure 2: Adder circuit configuration of task 2

The non-inverting voltage follower of task 3 consisted of only input voltage, output voltage measurement, and the operational amplifier. There were no resistances connected to any pins of the opamp. To achieve this setup we merely connected pin 2 with pin 6 of the opamp to form the feedback loop using one jumping wire, we set a signal driving the circuit through the non-inverting input.

Task 4 called for a differentiator circuit consisting of two capacitances of values $C = 10\text{nF}$, $C_p = 100\text{pF}$, and a resistance of value $R_v = 100\text{k}\Omega$. We later removed the capacitance C_p to determine the role it had in the differentiator circuit.

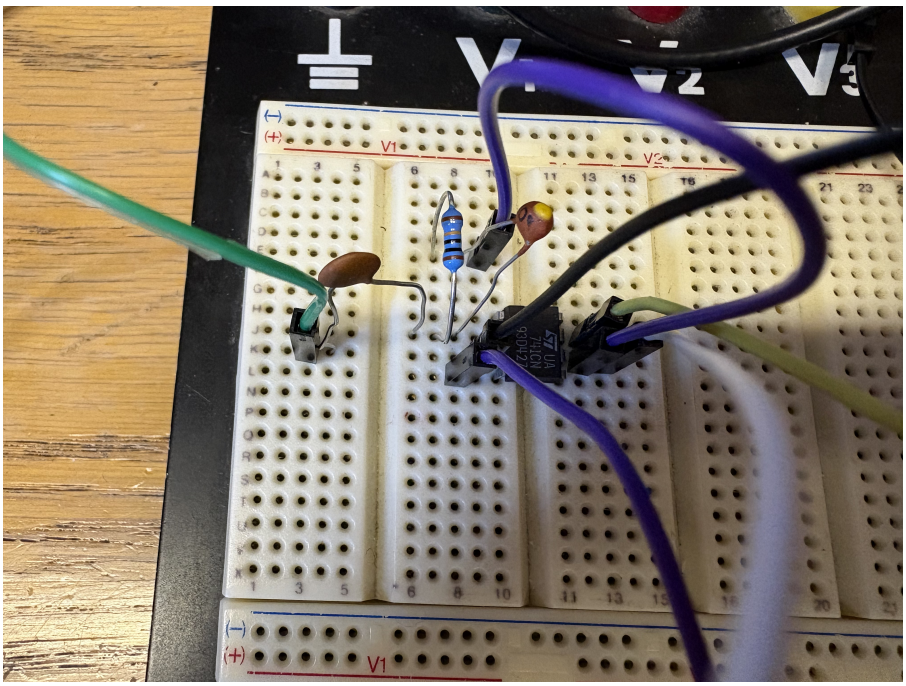


Figure 3: Differentiator circuit configuration of task 4

The integrator circuit was composed similar to the differentiator circuit with the minor change of removing C_p and reordering the resistances as $R_s = 100\text{k}\Omega$ and $R_P = 10\text{M}\Omega$. R_s was connected on the inverting input pin of the opamp while R_P was connected in parallel with C, both connected into the output pin of the opamp. The non-inverting input stayed grounded, as we drove the input voltage through the inverting input. As per the description of the task, we removed to R_P to examine the role it took within the integrator circuit.

4 Measurements

4.1 Saturation voltage of the inverting configuration

To test the inverting operational amplifier configuration of the μA741 opamp. We drove a DC input of 100 mV on the inverting input (pin 2) using the power supply instead of the signal generator. We chose this because while the signal generator can supply DC drive theoretically, our lab instructor mentioned that it is not a perfect DC signal and may result in some noise or unwanted artifacts.

Finding the saturation voltage consisted of increasing the input voltage with increments of 100 mV, and determining the position of the cutoff where the output voltage no longer shows an increase as the input is increased.

U_{in} [V]	U_{out} [V]
0.1	-0.578
0.2	-0.755
0.3	-1.145
0.4	-1.687
0.5	-2.241
0.6	-2.626
0.7	-3.059
0.8	-3.147
0.9	-3.147
1.0	-3.149
1.1	-3.149
1.2	-3.149

Table 1: Increasing DC input and output voltages of the inverting opamp configuration

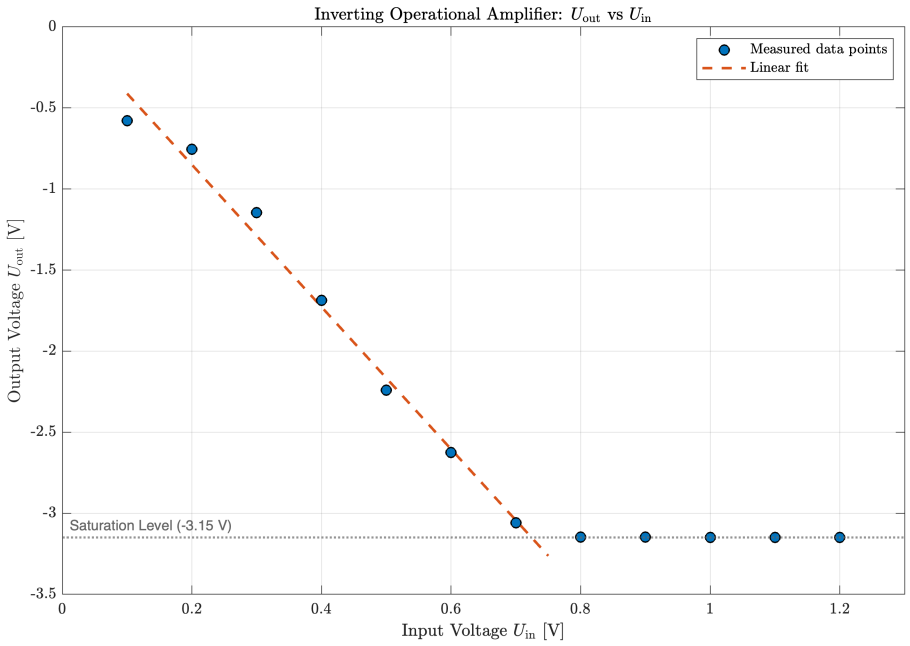


Figure 4: Finding the saturation voltage of the inverting opamp configuration with DC input

We found the saturation voltage to be at $U_{in,sat} = 800$ mV, where the output was $U_{out,sat} = -3.147$ V. From this point onwards the output voltage no longer increased with the same increment as previously, hence, here we determine the saturation voltage. Calculating the slope of the fitted line we found a closed loop gain of $A_u \approx -4.4$. Theoretically our gain, and hence, relative error should be:

$$A_u = -\frac{R_v}{R_s} = -\frac{470k\Omega}{100k\Omega} = -4.7$$

$$\text{Relative Error}(\%) = \frac{4.4-4.7}{4.7} \cdot 100\% \approx -6.38\%$$

To get a smaller relative deviation we could have measured the true resistance values of the resistors used with a multimeter, as they may have deviated from the theoretical value of 100 kΩ and 470 kΩ.

4.2 Operation of the adder amplifier

In order to study the operation of the adder amplifier, a circuit shown in the image below was built. Resistors with a value of 1 kΩ were used, 2 of them were placed in parallel and 1 was between the ground and the non-inverting input, and the two inputs of the circuit was connected. Three input voltages of 100 mV, 500 mV and 1 V were applied, and the corresponding output voltages were measured.

According to the equation provided in the Lab Description [1], the output voltage is proportional to the sum of the input voltages:

$$U_{out} = -\left(\frac{R_v}{R_1}U_{in1} + \frac{R_v}{R_2}U_{in2}\right). \tag{10}$$

Since $U_{in1} = U_{in2} = U_{in}$ and $R_v = R_1 = R_2$, the equation simplifies to $U_{out} = -2U_{in}$. The measured voltages along with the expected theoretical values are shown in Table 2.

Table 2: Measured and theoretical output voltages of the adder amplifier.

U_{in} (V)	$U_{out,meas}$ (V)	$U_{out,theor}$ (V)	Error (%)
0.100	-0.171	-0.200	14.50
0.500	-0.913	-1.000	8.70
1.000	-1.917	-2.000	4.15

The measured values are close to the expected theoretical numbers, and all the output voltages are negative, which confirms that the circuit has inverting behavior. As the voltage increases, the relative errors decrease from 14.5% to 4.15%.

4.3 Non-inverting voltage follower

A non-inverting voltage follower's operation was investigated using a sinusoidal input signal with 1 kHz frequency and 200 mV peak-to-peak voltage. Symmetrical voltages between +5V and -5V were applied to the operational amplifier. No noticable distortion of the waveform was observed at the cutoff frequency.

The meausred output voltage was 198 mV, from which one can determine the gain:

$$A_u = \frac{U_{out}}{U_{in}} = \frac{198}{200} = 0.99$$

Since the theoretical gain is $A_u = 1$, the obtained gain has a relative deviation of 1%.

The upper cutoff frequency can be determined by increasing the input frequency until the output amplitude decreases to $\frac{1}{\sqrt{(2)}}$ of the maximum value, while the waveform does not distort. Based on the measured output voltage of 198 mV, the calculated amplitude is:
$U_{out} = \frac{198}{\sqrt{2}} = 140$ mV. This output amplitude was observed at $f_c = 731$ kHz, which is the cutoff frequency determined experimentally.

4.4 Operation of the differentiator circuit

A differentiator circuit with a 10 nF capacitor and a $100k\Omega$ feedback resistor was assembled to examine its operation using sinusoidal, square and triangulas signals with 1kHz frequency. The effect of an additional 100 pF capacitor when connected in parallel with the feedback resistor was also studied. Instead of the 1 V input volage specified in the Lab Description, the measurement was performed using 500 mV, following the instructions of the laboratory supervisor (Farkas Viktor), since an applied input voltage of 1 V resulted in an output of approximately 6 V. The output voltage of the differentiator is calculated according to Equation 7.

A circuit accurately performs differentiation if the time constant is much smaller than the period of the signal to be differentiated. However, in the circuit investigated, $\tau = 100k\Omega \cdot 10nF = 1ms$ and the applied frequency was 1 kHz, therefore: $T = \frac{1}{f} = 1ms$, so $\tau = T$. Because of this, deviations from the ideal differentiation case are expected. According to theory, the expected waveforms for sinusoidal, square and triangle waveforms are a negative cosine wave, alternating positive and negative peaks at transitions, and an approximately square wave with inverted slopes, respectively. For a sinusoidal input, ideal inverting differentiation produces a phase shift of 270° [2]. Our experimentally obtained results are summarized in the following table:

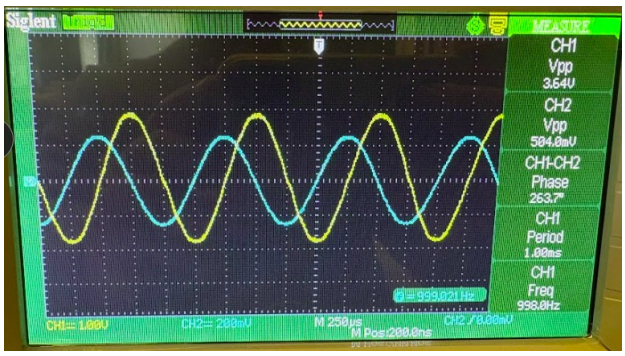
Table 3: Measured results of the differentiator circuit with and without C_P.

C_P	Input signal	U_{in} (V)	U_{out} (V)	Phase shift (°)
With	Sine	0.5	3.64	260
With	Square	0.5	7.68	177.4
With	Triangle	0.5	3.84	260
With	Triangle	1.8	7.80	235
Without	Sine	0.5	3.84	270
Without	Square	0.5	7.68	292
Without	Triangle	0.5	5.12	258

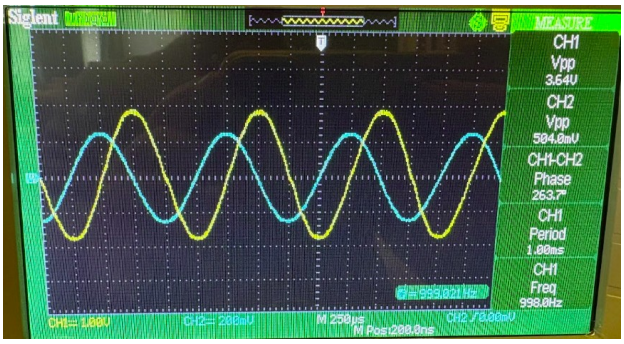
For the sinusoidal waveform the output voltages were 3.64 V and 3.84 V with and without C_P respectively. The corresponding phase shifts were 260° and 270°, which are close to the theoretical phase shift for an ideal inverting differentiator.

For the square wave, the measured output voltage was 7.68 V both with and without the additional capacitor. For the triangular input at 500 mV, the output voltages were 3.84 V and 5.12 V with and without capacitor, respectively. With the additional capacitor present, the output did not show a well-defifen rectangular waveform. An additional measurement was performed at an input voltage of 1.8 V, which reuslted in an output voltage of 7.8 V and a more rectangular-like waveform.

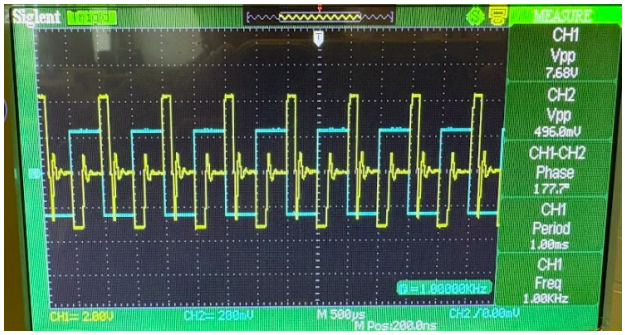
The additional capacitor connected in parallel with the feedback resistor influences the output amplitudes and phase shifts, as it limits the high-frequency response of the circuit.



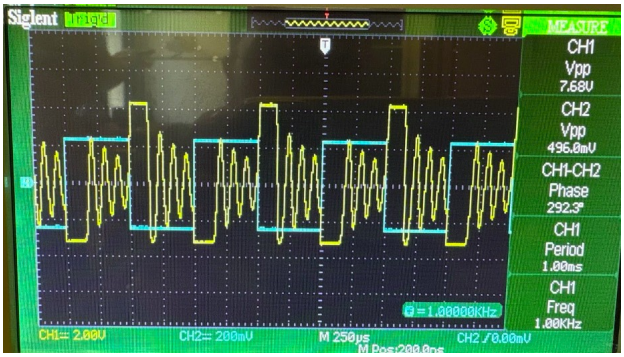
(a) Sine, with C_P



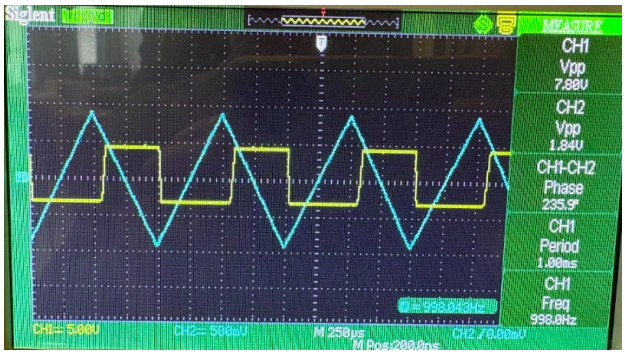
(b) Sine, without C_P



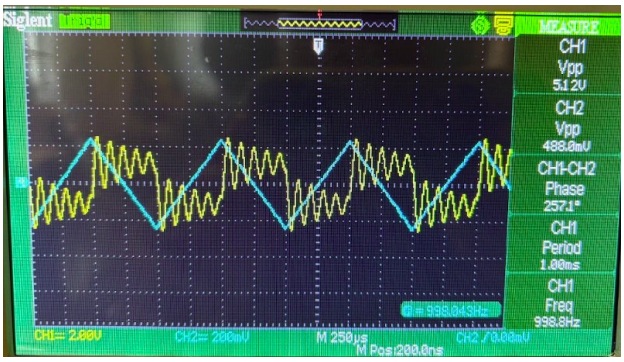
(c) Square, with C_P



(d) Square, without C_P



(e) Triangle, with C_P



(f) Triangle, without C_P

Figure 5: Oscilloscope measurements of the differentiator circuit for sinusoidal, square and triangular input signals, with and without the additional capacitor C_P.

As shown in the figure above, the measured waveforms approximately follow the expected behavior of the differentiator circuit. In the sinusoidal input case, a phase shift is observed. For the square-wave, peaks appear and in the triangular input case, the output is close to a rectangular waveform. The presence of the additional capacitor reduces the fluctuations and yields smoother output waveforms.

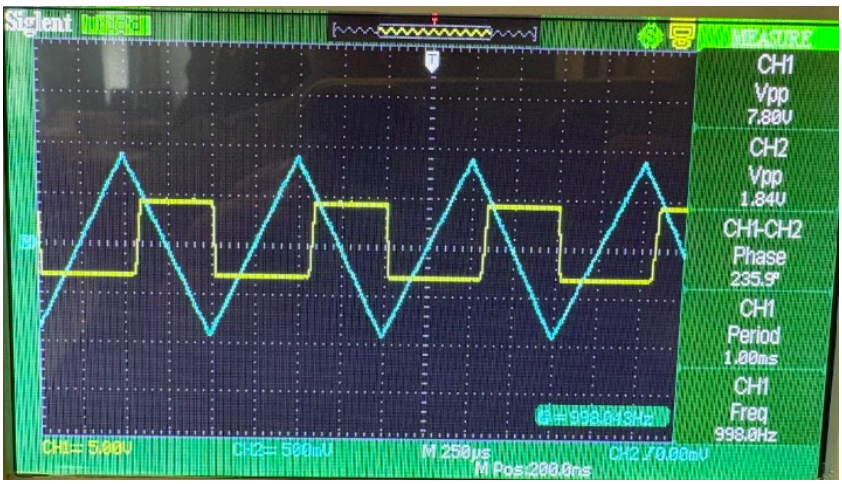


Figure 6: Additional oscilloscope measurement of the differentiator circuit with C_P for a triangular input signal of 1.8 V at 1 kHz.

As shown in the figure above, increased triangular input voltage (1.8 V) resulted in a more defined rectangular output.

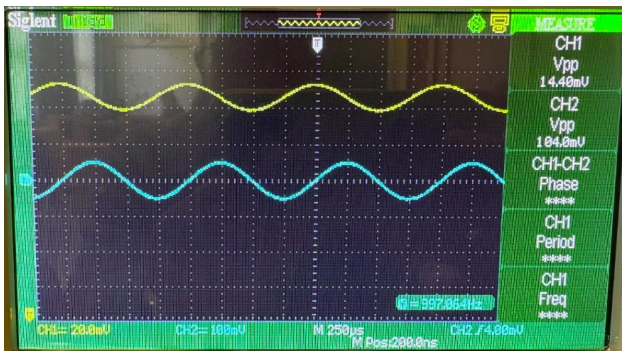
4.5 Operation of the integrator circuit

The operation of an integrator circuit was investigated with sinusoidal, square and triangular input signals with a frequency of 1 kHz and an input voltage of 100 mV. According to Equation 8, for a sine input wave in an integrator circuit, the expected output is a cosine wave. As for square and triangular waveforms, the expected output waveforms are triangular wave with inverted slope and parabolic waveform, respectively. In the assembled circuit, $\tau = R_P \cdot C = 10M\Omega \cdot 10nF = 0.1s$ and $T = \frac{1}{1000} = 0.001s$, therefore $tau = 100T >> T$, which meets the given condition[1].

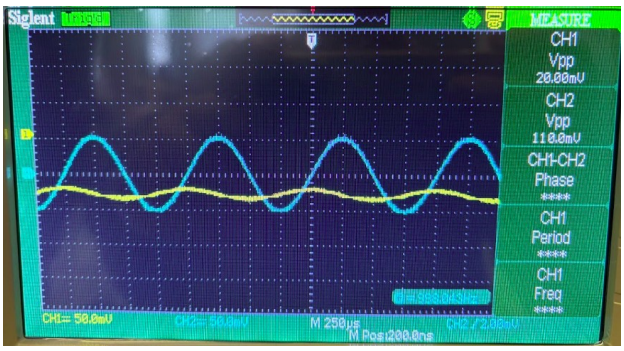
Table 4: Measured output voltages of the integrator circuit with and without R_P.

Input signal	With R_P (mV)	Without R_P (mV)
Sine	14.4	20.0
Square	22.4	28.0
Triangle	11.2	16.0

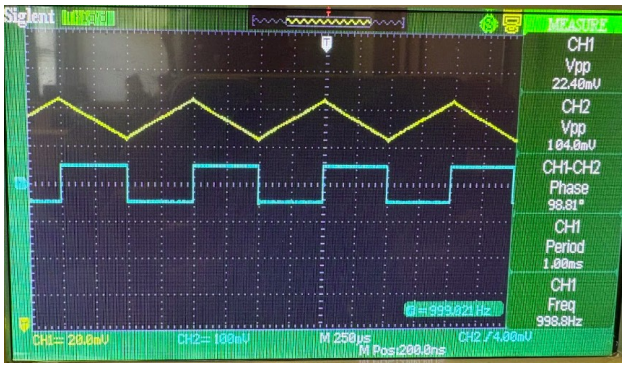
The measured output voltages are presented in the table above. With the additional resistor present in the circuit, the output voltages were 14.4 mV, 22.4 mV and 11.2 mV for the sinusoidal, square and triangular inputs, respectively. Without the additional resistor, the output voltages experienced an increase in their values: to 20 mV, 28 mV and 16 mV.



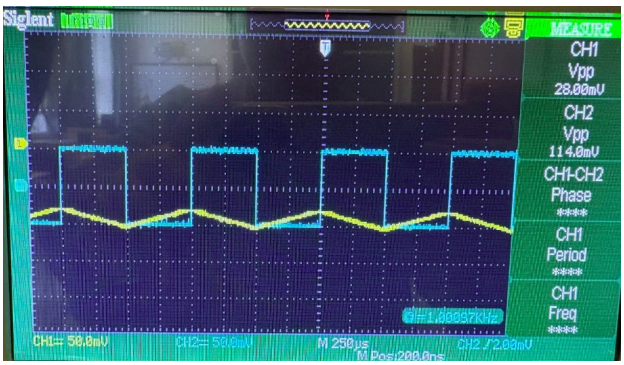
(a) Sine, with R_P



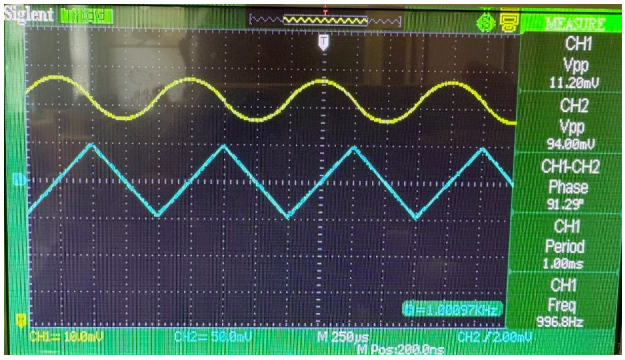
(b) Sine, without R_P



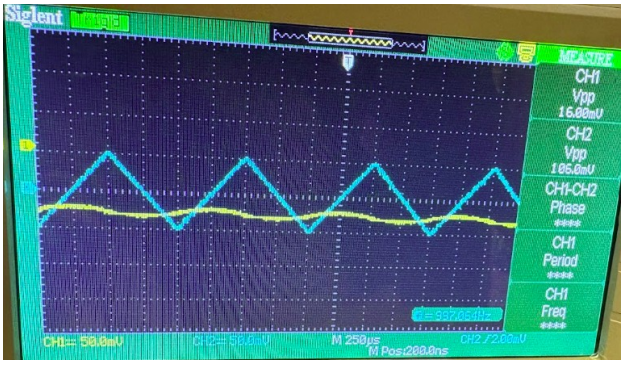
(c) Square, with R_P



(d) Square, without R_P



(e) Triangle, with R_P



(f) Triangle, without R_P

Figure 7: Oscilloscope measurements of the integrator circuit for sinusoidal, square and triangular input signals, with and without the additional resistor R_P.

As shown in Figure 7, the measured waveforms followed expectation. For the sinusoidal input, the output remained approximately sinusoidal, while acquiring a phase shift. In the square-wave input case, an approximately triangular output waveform was observed, and the triangular input led to a curved output waveform.

The measured output amplitudes were lower with R_P present and they were more clearly distinguishable. The additional resistor provides a DC feedback path, and as a consequence, it reduces the output drift and prevents the accumulation of DC errors [3].

5 Conclusion

The different configurations (inverting, adder, non-inverting voltage follower, differentiator, integrator) of operational amplifiers were successfully investigated during the laboratory session.

The measured voltage gains of the inverting amplifier, adder amplifier and voltage follower were close to the theoretical values. A saturating behavior for the inverting amplifier was observed and an upper cutoff frequency of 731 kHz was determined.

The measured waveforms for the differentiator and integrator circuits showed expected results, however, some deviation from the ideal response was observed.

Overall, the measurements demonstrated the operation of different amplifier configurations and the influence of various circuit components on the output.

References

[1] Viktor Farkas. *M2 - Basic Connections for Operational Amplifiers Lab Guide*. Budapest University of Technology and Economics. 2026.

[2] K. Webb. *Section 7: Active Filters*. ENGR 203 – Electrical Fundamentals III, Lecture Notes. Slide 15: Opamp Differentiator – Frequency Response. URL: `https://web.engr.oregonstate.edu/~webbky/ENGR203_files/Section%207%20Active%20FIlters.pdf`.

[3] Pete Semig. *Integrator Circuit*. Tech. rep. SBOA275B. Texas Instruments, 2024. URL: `https://www.ti.com/lit/an/sboa275b/sboa275b.pdf`.